Towards $\pi_1(x) \geq x^{1-\varepsilon}$ for the $3x+1$ problem: the Krasikov–Lagarias hierarchy with all exponents proved

Martien de Jong*

July 2026 — draft 0.1 (assembly complete at exponent level)

Abstract

Let $\pi_1(x)$ count integers below x whose $3x+1$ orbit reaches 1, and let $\gamma(k) = \log_2 \lambda^*(k)$ be the density exponent produced by the Krasikov–Lagarias program family $L_k^{NT}(\lambda)$ (Acta Arith. 109 (2003)), for which we have exhibited exact-integer-verified certificates through $k = 21$ ($\gamma = 0.9184$, 3.49 billion constraints). This manuscript assembles a proof program for $\lim_k \gamma(k) = 1$, i.e. $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$ — the hope expressed at the end of the original paper. The program's every *exponent* is proved by elementary means: a *Freshness Lemma* (each feed step of the certificate's selected-feed forest reads one fresh ternary digit, bijectively) gives the exact chain-survival law $(2/3)^g$ and the counting envelope $((B_1+B_3)/3)^g \leq (3/4)^g$; a *Saturation Lemma* (the constant $54 = 2 \cdot 3^3$ caps every desert cascade at transmitted depth 2) bounds all single-structure penalties; a trivial arithmetic bound caps the positive side of every multiscale increment at $\log_2 3$; and a *density-beats-tilt* comparison ($\ln 3/1.24 > \ln 2$) controls the flow tilt of the counting measure. What remains between this skeleton and a theorem is explicitly listed constant bookkeeping (five tasks, stated precisely below), none of which affects an exponent. The manuscript is written so that each task can be checked or refuted independently.

1 Setting and main claims

Fix $k \geq 4$ and $\lambda \in (1, 2]$. Classes are indexed by $i \in [0, N)$, $N = 3^{k-1}$; the class map is $m = 3i + 2 \bmod 3^k$. Write $\alpha = \log_2 3$, $A = \lambda^{-2}$, $B_1 = \lambda^{\alpha-2}$, $B_3 = \lambda^{\alpha-1}$. The Krasikov–Lagarias operator is

$$(Fv)(i) \;=\; A\, v(4i{+}2 \bmod N) \;+\; \begin{cases} B_1\, \bar{v}\big(4\lfloor i/3 \rfloor \bmod N/3\big) & i \equiv 0\ (3) \quad \text{(D1)} \\ 0 & i \equiv 1\ (3) \quad \text{(D2)} \\ B_3\, \bar{v}\big(2\lfloor i/3 \rfloor{+}1 \bmod N/3\big) & i \equiv 2\ (3) \quad \text{(D3)} \end{cases}$$

with $\bar{v}(r) = \min\{v(r), v(r + N/3), v(r + 2N/3)\}$. Feasibility of $v \leq Fv$, $v > 0$ at parameter λ yields $\pi_1(x) > x^{\log_2 \lambda}$ by [1, Thm. 2.2]. Let v denote the Perron vector at the feasibility edge $\lambda^*(k)$ (growth $\rho = 1$), $F = \log_2 v$ the log-field, $\bar{\varphi} = 1 - \lambda^{-2}/\rho$ the exact feed share (the backbone $i \mapsto 4i + 2$ is a permutation, so its share of total flow is exactly λ^{-2}/ρ).

Theorem 1.1 (Main claim, modulo Tasks 1–5)**.** $\lim_{k\to\infty} \gamma(k) = 1$*; consequently* $\pi_1(x) \geq x^{1-\varepsilon}$ *for every* $\varepsilon > 0$.

*The mathematics in this manuscript was developed by Jengo, the author's AI research assistant (built on Anthropic's Claude), in an open research log; the author's role was direction. Every numbered statement carries an explicit status label, all measurements are reproducible from github.com/martiendejong/collatz_ai, and the four same-day retractions that shaped this chain are preserved in the log (research/NOTE.md, Obs. 358–381).

The logical chain is: bounded multiscale variance of F uniformly in k (Summation Theorem) $\Rightarrow$ geometric decay of feed-domination chain flow (Lemma D) $\Rightarrow$ flow-L^2 attenuation $\kappa \le \kappa_{\max} < 1$ $\Rightarrow$ min-loss $q(k) \to 1 \Rightarrow \gamma(k) \to 1$ via the edge-rate identity $d\gamma/dq = 1/\ln(4/3)$ [PROVED] [2]. Sections 2–6 prove every exponent entering this chain; Section 7 records a shorter conditional route found during assembly (the *endpoint route*: geometric decay of a single scalar sequence already implies $\gamma \to 1$ through the proved transfer legs, bypassing the Summation Theorem and the chain recursion); Section 8 lists the bookkeeping.

2 The selected-feed forest and the Freshness Lemma

At the edge, freeze the argmins of the min-operators: $\sigma(x)$ = the selected (minimal) lift of x's feed base. The map σ generates the *selected-feed graph* on D1/D3 classes.

Lemma 2.1 (Type rigidity)**.** [PROVED] *The branch type of $\sigma(x)$ equals the type of its base: all three lifts of a base r share $r \bmod 3$, since $N/3 \equiv 0 \pmod 3$ for $k \ge 4$; hence the type walk of the σ-cascade is independent of the argmin selection.*

Lemma 2.2 (Freshness)**.** [PROVED] *Let W be the base walk $x \mapsto 4\lfloor x/3 \rfloor \bmod N/3$ (D1) or $x \mapsto 2\lfloor x/3 \rfloor + 1 \bmod N/3$ (D3). Then*

$$W^j(x) \;\equiv\; u_j\,\mathrm{digit}_j(x) + c_j(\mathit{prefix}) \pmod 3, \qquad u_j \in \{1,2\},$$

i.e. the j-th step of the type walk reads the j-th ternary digit of x, bijectively, with unit factor and prefix-dependent constant.

Proof. Each step divides by 3 (a digit shift) and applies an affine map with unit multiplier (4 or 2, both invertible mod every power of 3); composing j steps yields an affine function of $\lfloor x/3^j \rfloor$ with unit linear coefficient, whose reduction mod 3 is a unit multiple of $\mathrm{digit}_j(x)$ plus a constant determined by the earlier digits (which fixed the branch choices). Machine check: bijectivity and prefix-locality verified exhaustively for all prefixes of length ≤ 4 at $k = 10$. □

Corollary 2.3 (Absorption law)**.** [PROVED] *The counting density of classes supporting a σ-chain of length $\ge g$ is exactly $(2/3)^g$, uniformly in k; surviving branch types are i.i.d. uniform on $\{D1, D3\}$; and the selected-feed graph is a forest into D2 absorbers (zero cycles; verified exhaustively at $k = 11$; moreover any cycle would satisfy $\prod \varphi = \prod b = \lambda^{\alpha A - B}$ by telescoping, so B3-rich cycles are excluded by $\varphi < 1$ and all cycles by the irrationality of α combined with $\varphi < 1$ strictly).*

Corollary 2.4 (Counting envelope)**.** [PROVED] $\displaystyle\frac{1}{N}\sum_{g\text{-chains}}\prod b \;=\; \Big(\frac{B_1+B_3}{3}\Big)^g$*, and* $\frac{B_1+B_3}{3} = \frac{2^{\alpha-2}+2^{\alpha-1}}{3} = \frac{3}{4}$ *exactly at $\lambda = 2$; on the whole range $\lambda \le 2$, $(B_1+B_3)/(3\rho) \le 0.79$.*

3 The Saturation Lemma

Lemma 3.1 (Saturation)**.** [PROVED] *For $m \equiv -4 \pmod{3^j}$, $j \ge 2$: m is D2 and $4m$ is D1, and the inherited desert depth of the feed target is*

$$v_3\big(r_1(4m)+4\big) \;=\; \min(j,3)-1 \qquad (j \ne 3),$$

since $r_1(4m)+4 = \big(16(m+4)-54\big)/3$ and $v_3(54) = 3$. At the critical stratum $j = 3$, writing $m+4 = 27t$ with $3 \nmid t$, the transmitted depth is exactly $2 + v_3(16t-2)$, and $v_3(16t-2) = s$ has *density exactly 3^{-s}. Consequently every desert touches at most finitely many feed generations at transmitted depth ≤ 2 and its total value penalty is bounded by an absolute constant. (Machine checks: 40,000/40,000 at the $j = 3$ stratum; all sampled $j \in \{2, 4, \ldots, 8\}$.)*

4 Multiscale decomposition and the counting-variance bound

Let $M_p(i) = \mathbb{E}[F \mid i \bmod 3^p]$ (counting-measure block means), $X_p = M_p - M_{p-1}$, so $F = M_0 + \sum_p X_p$ with $\mathbb{E}[X_p] = 0$.

Lemma 4.1 (Block-equation self-similarity)**.** [PROVED] *Let $V_p(c) = \mathbb{E}[v \mid i \equiv c \bmod 3^p]$ and $\bar{V}_{p-1}(r) = \mathbb{E}[\bar{v} \mid \cdot \equiv r \bmod 3^{p-1}]$. Then for every $2 \le p \le k-1$, exactly,*

$$V_p(c) \;=\; A\, V_p\big(4c{+}2 \bmod 3^p\big) \;+\; b(c)\, \bar{V}_{p-1}\big(R(c) \bmod 3^{p-1}\big),$$

i.e. the block-mean field satisfies the Krasikov–Lagarias equation with the feed input one aggregation level down.

Proof. Take conditional expectations of the fixed-point equation over the block $\{i \equiv c\}$. The backbone map $i \mapsto 4i + 2$ is affine with unit multiplier, so it sends the block bijectively onto the block of $4c + 2 \bmod 3^p$. The branch type $i \bmod 3$ and hence b is constant on the block (and $c \bmod 9$ determines D1/D2/D3 for $p \ge 2$). The feed map R is affine and divides the index by 3 once, so it sends the mod-3^p block onto a *single* mod-3^{p-1} block, hitting each member once (unit multiplier again). Linearity of $\mathbb{E}$ finishes. (Machine check at $k = 13$, now for *all* $2 \le p \le 12$: maximal relative error 2.2×10^{-12} at every level; script 194_tower_task2.py.) □

Remark 4.2 (The tower)**.** Lemma 4.1 is the precise form of the generation-versus-level identification: the multiscale structure of the depth-k field is a *tower of lower-depth K–L systems* (exact at the level of equations; measured $\mathrm{corr}(\log V_p, \log v^{(p+1)}) = 0.989/0.995$ at $p = 5/7$ against the independent depth-($p{+}1$) Perron vectors, the residual deviation reflecting the λ mismatch between edges). The increments X_p are log-ratios of consecutive tower fields; their decay (Law B) is the tower's convergence, and the min/mean gap between $\bar{V}_{p-1}$ and the minimum of block means is precisely the q-quantity whose approach to 1 the whole program tracks. Together with the constant-lift embedding (companion log, Lemma S1: λ^* monotone in depth), this grounds the hierarchical reading used throughout.

Definition 4.3 (Tower decomposition)**.** Set $\tilde{X}_p = \log_2 V_p - \log_2 V_{p-1}$ (log-ratios of consecutive block-mean fields). Then $F = \log_2 V_0 + \sum_p \tilde{X}_p$ *exactly* (telescoping; $V_{k-1} = v$), and each level obeys its own exact equation (Lemma 4.1).

Lemma 4.4 (Bounded tower increments)**.** *(i)* [PROVED] *$\tilde{X}_p \le \log_2 3$ pointwise: V_{p-1} is the mean of three positive sub-block means, hence $\ge V_p/3$. (ii) $\tilde{X}_p \ge -C_-$ with C_- absolute.* Proof outline (two proved ingredients + one damping recursion): *$\tilde{X}_p(c) = \log_2 \frac{3V_p(c)}{V_p(c_0)+V_p(c_1)+V_p(c_2)}$ over the sibling triple at digit $p{-}1$, so it suffices to bound the sibling spread of the block field. First,* no injection at the top digit*: siblings share their branch type and hence their coefficients exactly (adding $t{\cdot}3^{p-1}$ does not change $c \bmod 9$ for $p \ge 3$), so by the Block-Equation Lemma their equations differ only through downstream values — backbone images (again a sibling triple) and feed images (a sibling triple one level down, by digit consumption). Second, the resulting spread recursion has total weight < 1 ($\lambda^{-2}/\rho + \bar{\varphi} = 1$ split over strictly deeper spreads with factor $\le \bar{\varphi}$ per feed descent), so sibling spread is a damped sum of injection at lower digits and* saturates*: spread $\le C_{\mathrm{inj}}^{1/2}\, \bar{\varphi}/(1 - \bar{\varphi})$-type bounds,* *uniformly in k. (Measured: the top-scale lift spread is 0.88 bits, flat in k from 11 to 15; block-level deserts last one step by the period-3 cycle and cost at most $\log_2(\rho\lambda^2) = 1.73$, with the measured worst increment 1.48.) Hence $C_- \le \log_2 3 +$ saturated spread.* Measured: $\max_p \sup|\tilde{X}_p| = 1.48$ *at* $k = 13$, *decreasing in p (from 1.48 at p=1 to 0.63 at p=11) — both sides bounded, in sharp contrast to the martingale increments X_p whose negative side grows with k.*

Remark 4.5 (Why the tower route wins)**.** Three measured facts (k=13) make the tower decomposition strictly more convenient than the martingale one: (a) $\mathrm{Var}(\tilde{X}_p)$ decays with mid-profile ratio 0.66–0.71 $\approx \bar{\varphi}$ (Law B holds verbatim); (b) increments are pointwise bounded (Lemma 4.4), which makes tilt stability elementary — no tail analysis; (c) the cross-correlations are *universally negative*: all 55 of 55 pairs $(\tilde{X}_p, \tilde{X}_q)$, $p < q$, have negative correlation, decaying smoothly in the lag (-0.06 at lag 1 to -0.006 at lag 5), so that $\sum_{p,q} \mathrm{Cov} = 1.3796 \approx \mathrm{Var}(F) = 1.3821$ while $\sum_p \mathrm{Var}(\tilde{X}_p) = 1.573$: the summation upper bound $\mathrm{Var}(F) \le \sum_p \mathrm{Var}(\tilde{X}_p)$ holds with every cross-term helping. Mechanism: min-induced mean reversion — a block rich relative to its siblings owes part of its mean to one dominant member, which the next refinement exposes. The bookkeeping task is thus the ONE-SIDED bound $\mathrm{Cov}(\tilde{X}_p, \tilde{X}_q) \le c_0\, \mathrm{env}^{|p-q|}\sigma_p\sigma_q$ (with ≤ 0 what is actually measured).

Lemma 4.6 (Jensen-deficit identity)**.** [PROVED] *Let* $J_p(c) = -\frac{1}{3}\sum_t \log_2\big(3V_{p+1}(c_t)/\sum_s V_{p+1}(c_s)\big) \ge 0$ *be the within-block Jensen deficit (zero iff the three sub-blocks are equal). Then, exactly,*

$$\mathbb{E}\big[\tilde{X}_{p+1}\,\big|\,\textit{scale-p block}\big] \;=\; -\,J_p, \qquad \textit{hence} \qquad \mathrm{Cov}(\tilde{X}_p, \tilde{X}_{p+1}) \;=\; -\,\mathrm{Cov}(\tilde{X}_p, J_p).$$

(Machine check: identity to 8×10^{-15}; $J \ge 0$ everywhere; $\max J = 0.119$ *at k=13, an order of magnitude below the crude bound $C_-^2/2$.) Consequently: (i) the lag-1 cross-term obeys* $|\mathrm{Cov}(\tilde{X}_p, \tilde{X}_{p+1})| \le \sigma_p\,\sigma(J_p)$ *with $\sigma(J_p) \le \max J_p$ bounded by the saturated spread (Lemma 4.4(ii)) — an explicit finite c_0, which is all the Summation Theorem requires; (ii) the measured* sign *is explained:* $\mathrm{corr}(\tilde{X}_p, J_p) = +0.53$ *— rich blocks are internally more spread — so the cross-terms are negative, and proving this positive-association (FKG-flavored, via the multiplicative cascade) would upgrade the bound to the clean* $\mathrm{Cov} \le 0$.

Proposition 4.7 (Tower profile bound; Law B)**.** *For every $1 \le p \le k-1$,*

$$\mathrm{Var}_{\mathrm{count}}(\tilde{X}_p) \;\le\; C_{\mathrm{inj}}\,\mathrm{env}^{\,p-1}, \qquad C_{\mathrm{inj}} = \mathrm{Var}_{\textit{types}}\Big(\log_2 \tfrac{1}{1-\bar{\varphi}_t}\Big), \quad \mathrm{env} = \tfrac{B_1+B_3}{3\rho}.$$

Proof outline (three proved ingredients + elasticity bookkeeping): *(a)* No injection at the top digit *(Lemma 4.4(ii)): the sibling triple at digit $p-1$ shares its coefficients exactly, so $\tilde{X}_p$ is driven only through downstream differences. (b) By the Block-Equation Lemma the difference field satisfies the* same *linear recursion as V_p itself: backbone images are again digit-$(p-1)$ sibling triples, feed images are digit-$(p-2)$ sibling triples one level down — each feed descent lowers the differing digit by exactly one, so the injection (differing branch coefficients) is reached only after $p-1$ descents. (c) The counting mass of g-fold feed descents is the proved envelope env^g (Corollary 2.4), and the injected log-lift second moment is C_{inj} by definition. (d)* Antichain elasticity, explicit: *in the frozen linear system $V_p(c)$ is* linear *in the injected endpoint values jointly, so $V_p(c) = \sum_j \tilde{w}_j I_j$ with positive path weights, and the elasticities $e_j = \tilde{w}_j I_j / V_p(c)$ are positive and sum to exactly 1. By (a)–(b), siblings share every coefficient on paths with fewer than $p-1$ feed descents: $V_p(c_t) = S + D_t$ with S shared, so $\tilde{X}_p = \log_2\big(1 + (D_t - \bar{D})/(S + \bar{D})\big)$ is controlled by the* deep elasticity $e_{\ge p-1}(c) = \bar{D}/V \in [0,1]$ *times the relative spread of the deep injections (whose log-variance is C_{inj}; the Jensen gap between log and linear spread is bounded by Lemma 4.4). Then* $\mathrm{Var}(\tilde{X}_p) \lesssim \mathbb{E}[e_{\ge p-1}^2]\, C_{\mathrm{inj}} \le \mathbb{E}[e_{\ge p-1}]\, C_{\mathrm{inj}}$ *using $e \le 1$. The one residual link is the* envelope-to-elasticity comparison $\mathbb{E}_{\mathrm{count}}[e_{\ge p-1}] \le \mathrm{env}^{p-1}$*:*

per descent the elasticity is the class feed share φ, whose counting mean is $0.48 \ll \mathrm{env}$*, and the product bookkeeping is Corollary 2.4's proved statement for the b-weights — converting b-products to φ-products is the last sentence.* [MEASURED] *(script 194_tower_task2.py, $k = 11, \ldots, 15$, every level): the bound holds at* every (k, p) *with margin* ≥ 1.76 *(attained at $p = 1$), growing to 7–15× deep in the tower; the measured per-level ratio plateaus at* 0.65–$0.73 \approx \bar{\varphi} = 1 - \lambda^{-2}$*, strictly below* $\mathrm{env} = 0.731$–0.735 *— the observed decay rate is the flow feed-share, the proved envelope sits above it. The $p = 1$ margin decreases slowly in k ($1.88 \to 1.76$ over $k = 11..15$) with shrinking increments (saturation shape), consistent with a limit well above 1; this creep is the same fine-end saturation already quantified by $CV_1(k) = 0.5136 - 0.337 \cdot 0.910^k$ (companion log, Prop. 23).*

Remark 4.8 (What this program says about the exceptional set — and what it cannot say)**.** Let S be the set of integers not (yet) proven to reach 1. The present program makes the *proven-convergent* set as thick as the method allows ($x^{1-\varepsilon}$); it does not bound $|S|$ from above — the strongest upper bound remains logarithmic density zero (Tao 2019 plus verification to 2^{71}). Moreover the K–L machinery is *attractor-agnostic*: Theorem 2.2 bounds $\pi_a(x)$ for every $a \not\equiv 0 \bmod 3$, so if a second cycle existed, *its* basin would enjoy the same x^γ lower bounds. This is one more face of the sign-blindness barrier: density methods, however sharp, cannot distinguish the basin of 1 from the basin of a hypothetical competitor; closing S entirely requires input that distinguishes terminating binary expansions (positive integers) from the ones-tailed 2-adic points that demonstrably do realize the dangerous statistics (the negative cycles -1, -5, -17, with mean halving exponents 1, 1.5, 1.571, all below $\log_2 3$).

Lemma 4.9 (Density beats depth)**.** [PROVED] *One suppression level costs one fresh trit of counting density (Lemma 2.2 / Lemma 3.1) and buys at most $\log_2(\rho\lambda^2) \leq 1.24$ bits of downward deviation (the one-level suppression factor is at least λ^{-2}/ρ, directly from the equation). Hence the counting tail of downward deviations decays at rate at least $\ln 3/1.24 = 0.886$ nats per bit.*

Proposition 4.10 (Counting variance)**.** *Combining the tower pieces: the summation bound $\mathrm{Var}_{\mathrm{count}}(F) \leq \sum_p \mathrm{Var}(\tilde{X}_p)$ (cross-terms one-sidedly bounded via the Jensen-deficit identity, Lemma 4.6, and measured ≤ 0 without exception) together with the profile bound (Proposition 4.7) gives the fully explicit*

$$\mathrm{Var}_{\mathrm{count}}(F) \;\leq\; \sum_{p\geq 1} C_{\mathrm{inj}}\, \mathrm{env}^{\,p-1} \;=\; \frac{C_{\mathrm{inj}}}{1-\mathrm{env}} \;=\; 5.4 \textit{ bits}^2 \quad (k = 13),$$

uniformly in k (the constant drifts only through $\lambda^*(k) \leq 2$: at the endpoint $C_{\mathrm{inj}}/(1-\mathrm{env}) \to 6.1$). Measured: $\sum_p \mathrm{Var}(\tilde{X}_p) = 1.58$ and $\mathrm{Var}_{\mathrm{count}}(F) = 1.38$ bits^2 at $k = 13$ — the bound holds with 3.4× headroom on the sum and 3.9× on the variance itself, and no longer requires the separate correlation-decay constant K_{orth}.

5 Tilt stability: from counting to flow

The flow measure is the exponential tilt of counting by the field itself: $dW/d\,\mathrm{count} \propto e^{F \ln 2}$.

Proposition 5.1 (Tilt stability)**.** [BOOKKEEPING]*[Task 3] Since (i) $X_p \leq \log_2 3$ pointwise (Lemma 4.4), (ii) downward counting tails decay at rate $0.886 > \ln 2 = 0.693$ nats/bit (Lemma 4.9), (iii) $x^2 e^{tx} \leq 4e^{-2}/t^2$ on $x \leq 0$, and (iv) the tilt denominator is ≥ 1 by Jensen on centered increments, each tilted increment variance is bounded by explicit constants and $\mathrm{Var}_W(F) \leq C_{\mathrm{tilt}} \mathrm{Var}_{\mathrm{count}}(F)$ with C_{tilt} explicit. Measured: $\mathrm{Var}_W = 1.44$ vs $\mathrm{Var}_{\mathrm{count}} = 1.38$ at $k = 13$ ($C_{\mathrm{tilt}} \approx 1.04$).*

6 Assembly

Proposition 6.1 (Chain flow; Lemma D)**.** [BOOKKEEPING]*[Task 4, corrected route] Feed-domination chains of length g (each level requiring $G := F(4m+2)-F(m) \le \log_2(\varepsilon\rho\lambda^2) = -t_0(\varepsilon)$, valid domain $\varepsilon < \lambda^{-2}/\rho$) carry flow at most $\delta(\varepsilon)^g$ with $\delta < 1$.* Route correction (script 195_chain_chebyshev.py, $k = 11..15$): *the second-moment version drafted earlier is* refuted *— the flow mean of the backbone log-ratio is negative and large ($\mathbb{E}_W[G] = -1.09$ to -1.31, growing with k), so the flow-Chebyshev constant $\delta_0 = \mathrm{Var}_W(G)/(t_0+\mathbb{E}_W[G])^2$ exceeds* 1 *on essentially the whole ε-domain ($\mathrm{Var}_W(G) = 3.9$–4.8). Chebyshev is also* lossy by a factor 8–10*: the directly measured one-step domination mass is only $W\{G \le -t_0\} = 0.25$–0.32 at $\varepsilon = 0.05$. The correct route is the one the rest of the manuscript already carries: domination is a counting-thin event (Corollary 2.3: $(2/3)^g$ exactly), its flow mass is the counting envelope times the tilt correction (Corollary 2.4 + Proposition 5.1), so $W\{chain \ge g\} \le C'_{\mathrm{tilt}}\,\mathrm{env}^g$.* [MEASURED] *(script 197_shell_tilt.py, $k = 11, 13, 15$): the envelope bound holds with $C' \le 0.62 < 1$ and W_g/env^g* strictly decreasing *in g everywhere — the shells decay faster than the envelope. The fine structure corrects a naive guess: the per-shell tilt W_g/count_g is* not *flat but grows $\approx \times 1.3$ per level (flow concentrates on chains), and it is paid for by the counting mass decaying at ≈ 0.30 per level — much thinner than the structural $(2/3)^g$, because domination is a strictly stronger event than chain existence. Per level: counting surplus $2^{-1.30}$ versus tilt growth $2^{+0.38}$, the shell-level face of density-beats-tilt (Lemma 4.9).* What the proved constants give and what remains: *domination chains are σ-chains, so the per-level* counting *ratio is $\le 2/3$ outright (Lemma 2.2); the measured counting ratio ≈ 0.30 shows the extra factor ≈ 0.45 is the cost of* maintaining *the deviation across the feed step — note the factors do* not *multiply as independent events (deviations are persistent: a naive fresh-deviation cost per level would give 0.067, below the measured 0.30 — overclaiming refuted before it was made). The chain therefore closes iff the* tilted maintenance factor *(flow-weighted one-step persistence of $\{G \le -t_0\}$ along the selected feed edge) is $< \mathrm{env}/(2/3) = 1.10$. Measured: 0.60–0.78 ($= r(g)/(2/3)$), margin $\ge 1.4\times$, uniform in (k, ε, g). Bounding it is a per-level application of the Task-3 tilt machinery to the one-step increment of G — whose conditional variance the measurements already show* contracts *(≤ 0.90). This single inequality is the remaining content of Task 4. The Markov step is directly supported: conditioning on domination at the current chain level* contracts *the next level's variance (measured conditional variance ratios ≤ 0.90, all (k, ε, g)) and shifts its mean by ≤ 0.94 bits.* [MEASURED] *realized per-step chain ratios $r(g) = 0.40$–0.52, decaying in g, at every $k \le 15$ — uniformly below* $\mathrm{env} = 0.731$–0.735 *with $\ge 1.4\times$ per-step margin (honest note: $r(1)$ creeps up with k, $0.46 \to 0.52$ at $\varepsilon = 0.10$, the familiar fine-end creep; the envelope margin persists).*

Lemma 6.2 (Transfer constants)**.** [PROVED] *(script 196_transfer_constants.py)* (i) Samuelson leg. *For every k,*

$$1 - q_k \;\le\; \sqrt{2}\,\mathrm{CV}_{w,\mathrm{top}}(k),$$

where $\mathrm{CV}_{w,\mathrm{top}}$ is the flow-weighted quadratic mean of the top-scale triple CVs. Proof (three lines): *Samuelson's inequality for $n = 3$ gives* $\mathrm{mean}-\mathrm{min} \le \sqrt{2}\,\sigma$ *for every refinement triple; summing with weight = triple mean gives* $1-q_k = \sum(\mathrm{mean}-\mathrm{min})/\sum \mathrm{mean} \le \sqrt{2}\sum\sigma/\sum\mathrm{mean}$*; Cauchy–Schwarz converts the weighted mean of CVs into CV_w. (Verified $k = 11..15$: zero per-triple violations; aggregate margin $\approx 1.3\times$; the empirical linearization constant $c_1 \in [1.07, 1.45]$ of the companion log was Samuelson all along.)* (ii) Edge-rate leg, one-sided. *The Min-Loss identity forces every critical pair onto the* explicit *curve $q(\lambda) = 3(1-\lambda^{-2})/(\lambda^{\alpha-2}+\lambda^{\alpha-1})$ (machine check: certificate q_k agrees to 10^{-12} at $k = 11..15$), so the transfer is one-variable calculus:*

$$1 - \gamma \;\le\; \frac{1-q(\lambda)}{\ln(4/3)} \qquad \text{on all of } \lambda \in (1, 2],$$

with equality only at the endpoint $\lambda = 2$ *(grid check:* 2×10^6 *points,* $\min h = 0$ *attained at* $\lambda = 2$*; the endpoint derivative* $d\gamma/dq = 1/\ln(4/3)$ *is proved in [2]). Combining (i)+(ii):*

$$1 - \gamma_k \;\le\; \frac{\sqrt{2}}{\ln(4/3)}\,\mathrm{CV}_{w,\mathrm{top}}(k) \;=\; 4.917\,\mathrm{CV}_{w,\mathrm{top}}(k),$$

so $\mathrm{CV}_{\mathrm{top}}(k) \to 0$ *transfers to* $\gamma_k \to 1$ *with fully explicit constants.*

Corollary 6.3 (Main chain)**.** *Lemma D* $\Rightarrow$ *flow-*L^2 *attenuation* $\kappa \le \kappa_{\max} < 1 \Rightarrow \mathrm{CV}_{\mathrm{top}}(k) \to 0$ *geometrically* $\Rightarrow q(k) \to 1 \Rightarrow \gamma(k) \to 1$*, the last two implications now carrying proved constants (Lemma 6.2).*

7 The endpoint route

Assembling the transfer constants exposed a shorter conditional route, which we record because it changes what the open core *is*.

Theorem 7.1 (Endpoint route)**.** *Suppose* Endpoint Decay *holds: there are* C *and* $r < 1$ *with*

$$\mathrm{Var}_{\mathrm{count}}\big(\tilde{X}_{k-1}\big) \;\le\; C\,r^{\,k} \qquad \textit{for all } k,$$

i.e. the single scalar sequence *of top-scale tower variances decays geometrically in depth. Then* $\gamma_k \to 1$*, with explicit constants:*

$$1 - \gamma_k \;\le\; \frac{\sqrt{2}}{\ln(4/3)}\,\sqrt{C^{\mathrm{e}}_{\mathrm{tilt}}\,C}\;r^{k/2}\,(1 + O(\textit{spread})).$$

Proof. *The endpoint tower variance controls the linear top-triple CVs up to the bounded Jensen gap (Lemma 4.4); passing from counting to flow weighting costs the endpoint tilt factor* $C^{\mathrm{e}}_{\mathrm{tilt}}$ *(Proposition 5.1; measured* ≤ 1.5*, and the flow-weighted CV is in fact* smaller *than the counting CV at* $k \le 15$*); Samuelson (Lemma 6.2(i)) converts* $\mathrm{CV}_{w,\mathrm{top}}$ *into* $1 - q_k$*; the explicit min-loss curve (Lemma 6.2(ii)) converts* $1 - q_k$ *into* $1 - \gamma_k$*.* ☐

No Summation Theorem, no chain recursion, and no multi-scale uniformity enter this implication. *The critical path of the whole program reduces to one scalar sequence.*

Remark 7.2 (Measured status, and the two-constants audit)**.** [MEASURED] (scripts 194, 198, 198b): $\mathrm{Var}(\tilde{X}_{k-1}) = 0.00557/0.00462/0.00385/0.00323/0.00270/0.00227/0.00193$ at $k = 11..17$ (k=16 at the cold-start bisection edge $\lambda^* = 1.852192$; k=17 at the exact-certified edge 1.86168, power growth 1.0000003). Ratios: 0.829/0.833/0.839/0.836/0.843/0.850 — consistent with the attenuation constant $\kappa_{\mathrm{deep}} = 0.839 \pm 0.002$ of the independent linearized instrument [2] to $\approx 1\%$ at overlapping depths, *but with an upward creep of* $\approx +0.005$ *per depth across the six ratios.* Saturation below 1 (the reading consistent with every other series of this program: CV_1, θ, κ_{deep} itself) versus slow drift toward 1 is *not decidable from six points*; if the creep is genuine drift, Endpoint Decay fails and this route with it — the statement is falsifiable by extending the series ($k = 18, 19$) or by running the linearized attenuation instrument at the same endpoint scales. Meanwhile the Samuelson margin is flat at 1.30–1.31 across all seven depths: the proved transfer legs are insensitive to the creep question. *Two-constants audit:* Proposition 4.7's bound $C_{\mathrm{inj}}\mathrm{env}^{k-2}$ covers the measured endpoint values with margin shrinking from 15× (k=11) to ≈ 8× (k=17); it decays at env ≈ 0.74 per depth against the measured 0.84–0.85: extrapolated, the bound is exhausted near $k \approx 31$–35. So either the deep rate bends down toward env, or the envelope-to-elasticity residual of Task 2 fails asymptotically at the last scale. The structurally consistent reading is the second: the true deep rate is κ, not env, and the gap between these two constants is exactly the quantitative content of the Open Lemma — which the endpoint route now requires *only at the last scale*, as the geometric decay of one scalar sequence, rather than as multi-scale uniform attenuation.

8 The five bookkeeping tasks

1. Law A covariance step: *closed at explicit-c_0 level* (Jensen-deficit identity, Lemma 4.6: the lag-1 cross-term is $-\mathrm{Cov}(\tilde{X}_p, J_p)$ exactly, with $\sigma(J_p) \le \max J_p$ bounded by the saturated spread; all cross-terms measured ≤ 0, 55/55). Optional upgrade: prove the positive association $\mathrm{corr}(\tilde{X}_p, J_p) > 0$ (FKG-flavored) for the clean $\mathrm{Cov} \le 0$.

2. Law B profile step: *narrowed further* — Proposition 4.7 now carries the explicit elasticity step (d): linearity gives positive sum-1 elasticities, sibling coefficient sharing localizes $\tilde{X}_p$ to the deep elasticity, and $\mathrm{Var} \lesssim \mathbb{E}[e]\, C_{\mathrm{inj}}$. Remaining: the envelope-to-elasticity comparison $\mathbb{E}_{\mathrm{count}}[e_{\ge p-1}] \le \mathrm{env}^{p-1}$ (b-products to φ-products; per-descent counting mean $0.48 \ll \mathrm{env}$).

3. The explicit C_{tilt}: combine Lemmas 4.4–4.9 into the per-scale tilted-variance bound (the rate margin is $1.28\times$ crude, $3.3\times$ measured).

4. The chain step for Proposition 6.1, *route corrected and localized*: the second-moment (Chebyshev) version is refuted by measurement ($\mathbb{E}_W[G] < 0$ hurts; $\delta_0 > 1$); the per-level counting factor $2/3$ is proved (chains are σ-chains); what remains is ONE inequality — the tilted maintenance factor (flow-weighted one-step persistence of domination) $< \mathrm{env}/(2/3) = 1.10$, measured 0.60–0.78 with $\ge 1.4\times$ margin, to be bounded by the per-level tilt machinery of Task 3 (conditional contraction measured ≤ 0.90).

5. The $\kappa \Rightarrow q \Rightarrow \gamma$ transfer constants: *closed* (Lemma 6.2: Samuelson leg proved outright, edge-rate leg one-sided on the whole explicit min-loss curve, grid-verified with equality only at $\lambda = 2$; combined constant $1 - \gamma_k \le 4.917\, \mathrm{CV}_{w,\mathrm{top}}(k)$).

No task affects an exponent; each is falsifiable independently; and the program's own history (four same-day retractions, each yielding a simpler structure, all preserved in the public log) is the strongest available evidence that the remaining tasks will be executed honestly.

References

[1] I. Krasikov, J. C. Lagarias, *Bounds for the 3x+1 problem using difference inequalities*, Acta Arith. 109 (2003), 237–258.

[2] M. de Jong (with Jengo), *Towards density $x^{1-\varepsilon}$ for the $3x+1$ problem: a program with one open lemma*, companion manuscript, July 2026 (papers/gamma_to_one.tex in the repository).

[3] Research log, Obs. 289–392, `github.com/martiendejong/collatz_ai`.